

Supplementary Material: Leakage-Controlled Concept Graph Construction and Cold-Start Diagnostic Protocol for Knowledge Tracing

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Abstract

This supplement provides per-dataset baseline and ablation tables, paired significance tests, full cold-start strata, DDR downstream scatter plot, sequence-autocorrelation statistics, operator definitions, ground-truth disagreement examples, extended ground-truth metrics with node Jaccard, and supplementary DDR / KT-graph figures omitted from the main manuscript for length. Tables S15–S18 report training parity, Δ AUC intervals (paired- t by default), controlled leak-injection metrics, and downstream AUC on injected graphs. Table S21 reports the epoch-matched GKT ablation on XES3G5M.

Supplementary index

Tables. S1–S5: per-dataset baseline diagnostics (Section 1); S6–S10: per-dataset graph ablation (Section 2); S11: paired significance tests on public benchmarks (Section 3); S12–S13: cold-start diagnostics (Section 8); S14: ground-truth metrics including node Jaccard (Section 7); S15: training parity (Section 11); S16: Δ AUC intervals for primary pairs (Section 12); S17–S18: controlled leak injection and downstream AUC

Table 1: Diagnostic baseline results (train-only graphs). Baselines are used for protocol comparison only; no SOTA claim is made. Values are fold means with 95% bootstrap confidence intervals over valid+test sequence positions (`n_folds` as exported per dataset).

Model	AUC		ACC		NLL	
AKT	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.121	[0.093, 0.135]
BKT	0.780	[0.779, 0.781]	0.846	[0.845, 0.847]	0.379	[0.377, 0.380]
DGEKT	0.981	[0.981, 0.981]	0.943	[0.942, 0.943]	0.137	[0.136, 0.138]
DKT	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.120	[0.093, 0.134]
DyGKT	0.978	[0.978, 0.978]	0.934	[0.933, 0.934]	0.151	[0.151, 0.152]
GIKT	0.982	[0.982, 0.982]	0.946	[0.946, 0.946]	0.133	[0.132, 0.133]
<i>simpleKT</i>	0.984	[0.982, 0.989]	0.951	[0.945, 0.962]	0.121	[0.093, 0.135]
SKT	0.979	[0.976, 0.986]	0.938	[0.931, 0.954]	0.139	[0.107, 0.156]

Table 2: Diagnostic baselines: ASSISTments 2012 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.968	[0.968, 0.969]	0.916	[0.914, 0.917]	0.175	[0.173, 0.176]
BKT	0.669	[0.667, 0.671]	0.713	[0.711, 0.715]	0.574	[0.573, 0.576]
DGEKT	0.960	[0.959, 0.961]	0.907	[0.906, 0.908]	0.186	[0.185, 0.188]
DKT	0.963	[0.961, 0.964]	0.910	[0.908, 0.911]	0.182	[0.180, 0.185]
DyGKT	0.952	[0.951, 0.952]	0.899	[0.898, 0.900]	0.200	[0.198, 0.203]
GIKT	0.967	[0.967, 0.968]	0.915	[0.914, 0.915]	0.175	[0.174, 0.177]
GKT	0.961	[0.960, 0.962]	0.907	[0.906, 0.908]	0.185	[0.184, 0.187]
<i>simpleKT</i>	0.968	[0.967, 0.968]	0.916	[0.915, 0.916]	0.173	[0.172, 0.175]
SKT	0.953	[0.952, 0.953]	0.898	[0.897, 0.899]	0.198	[0.196, 0.200]

(Section 14); S19–S20: exploratory ANOVA on backbone and DGEKT downstream AUC (Section 15); S21: epoch-matched GKT ablation (Section 13).

Sections and figures. Section S3: sequence-autocorrelation analysis (Section 4); Section S4: DDR augmentation operators (Section 5); Section S5: ground-truth disagreement examples (Section 6); Figure S1: multi-benchmark train-only E_{pre} samples (Figure 3); Figure S2: multi-benchmark DDR curves (Figure 4); Figure S3: DDR downstream scatter plot (Figure 2). Cold-start panels in the main manuscript (Figure 7) are generated from fold 0 exports by `scripts/plot\_kt\_graph\_figures.py`.

1 Per-dataset baseline diagnostics

Tables S1–S5 (Tables 1–5) reproduce the five diagnostic baseline tables (AUC, ACC, NLL with 95% bootstrap CIs) that the main manuscript summarises in its cross-dataset AUC summary table. pyKT GKT is omitted on Junyi Academy because the stock wrapper is impractical on the dense multi-million-interaction graphs under our compute budget.

Table 3: Diagnostic baselines: XES3G5M (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.875	[0.874, 0.875]	0.858	[0.857, 0.858]	0.324	[0.323, 0.326]
BKT	0.636	[0.634, 0.638]	0.793	[0.791, 0.793]	0.490	[0.490, 0.492]
DGEKT	0.841	[0.841, 0.842]	0.844	[0.844, 0.845]	0.358	[0.356, 0.358]
DKT	0.858	[0.857, 0.858]	0.851	[0.851, 0.852]	0.342	[0.341, 0.342]
DyGKT	0.752	[0.752, 0.754]	0.813	[0.813, 0.814]	0.431	[0.429, 0.431]
GIKT	0.878	[0.878, 0.878]	0.860	[0.859, 0.860]	0.321	[0.320, 0.322]
GKT	0.834	[0.833, 0.835]	0.842	[0.841, 0.842]	0.364	[0.363, 0.365]
<i>simpleKT</i>	0.875	[0.874, 0.875]	0.858	[0.857, 0.859]	0.324	[0.323, 0.325]
SKT	0.625	[0.623, 0.626]	0.800	[0.799, 0.801]	0.482	[0.480, 0.483]

Table 4: Diagnostic baselines: Synthetic C2 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.873	[0.870, 0.876]	0.808	[0.802, 0.812]	0.407	[0.401, 0.417]
BKT	0.835	[0.833, 0.838]	0.782	[0.780, 0.784]	0.460	[0.458, 0.462]
DGEKT	0.710	[0.634, 0.859]	0.724	[0.686, 0.794]	0.551	[0.432, 0.613]
DKT	0.862	[0.857, 0.868]	0.798	[0.788, 0.805]	0.421	[0.413, 0.434]
DyGKT	0.710	[0.631, 0.867]	0.726	[0.686, 0.801]	0.550	[0.421, 0.617]
GIKT	0.875	[0.871, 0.880]	0.809	[0.802, 0.815]	0.405	[0.398, 0.416]
GKT	0.873	[0.868, 0.878]	0.808	[0.799, 0.814]	0.407	[0.400, 0.420]
<i>simpleKT</i>	0.875	[0.871, 0.880]	0.809	[0.803, 0.814]	0.405	[0.399, 0.416]
SKT	0.708	[0.634, 0.853]	0.722	[0.686, 0.788]	0.554	[0.440, 0.613]

2 Per-dataset graph construction ablation

Tables S6–S10 (Table 6–10) report fold 0 train-only versus full-log metrics for each graph-augmented pyKT model. The main manuscript reports only the maximum $|\Delta\text{AUC}|$ and $|\Delta\text{ACC}|$ per dataset.

3 Paired significance tests (public benchmarks)

Table S11 (Table 11) reports paired t -test and Wilcoxon p -values for graph-augmented models versus *simpleKT* on the three public benchmarks. The main manuscript baseline-CV table lists only Wilcoxon p -values because three-fold CV limits Wilcoxon resolution ($p_{\min}=0.25$).

4 Sequence autocorrelation analysis

For every learner we order interactions by timestamp and, over consecutive within-learner pairs $(t-1, t)$, measure KC-repeat rate, item-repeat rate, label persistence, lag-one label autocorrelation ρ , and the AUC of the trivial previous-answer predictor (`scripts/compute_autocorrelation.py`).

Table 5: Diagnostic baselines: Synthetic C5 (three-fold learner CV (`n_folds=3`); train-only graphs).

Model	AUC		ACC		NLL	
AKT	0.822	[0.819, 0.823]	0.750	[0.746, 0.752]	0.500	[0.496, 0.507]
BKT	0.791	[0.789, 0.794]	0.724	[0.722, 0.726]	0.534	[0.532, 0.536]
DGEKT	0.830	[0.827, 0.832]	0.756	[0.755, 0.757]	0.489	[0.486, 0.492]
DKT	0.826	[0.825, 0.827]	0.754	[0.752, 0.756]	0.493	[0.491, 0.496]
DyGKT	0.826	[0.824, 0.828]	0.751	[0.748, 0.753]	0.492	[0.490, 0.496]
GIKT	0.820	[0.817, 0.824]	0.749	[0.744, 0.754]	0.500	[0.496, 0.504]
GKT	0.810	[0.806, 0.814]	0.738	[0.730, 0.744]	0.526	[0.513, 0.538]
<i>simpleKT</i>	0.829	[0.828, 0.831]	0.755	[0.754, 0.757]	0.491	[0.489, 0.492]
SKT	0.810	[0.808, 0.813]	0.740	[0.737, 0.744]	0.511	[0.507, 0.513]

Table 6: Ablation contrasting *train-only* graph smoothing (protocol) with a *full-log* graph built from all learners before splitting (leaky construction). Graph-augmented rows are pyKT checkpoints consuming exported adjacency from P0. Δ is fold 0 full-log minus train-only. *to*: train-only graph; *fl*: full-log graph.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GIKT	0.982	0.982	0.000	0.946	0.946	-0.000
SKT	0.979	0.976	-0.003	0.938	0.931	-0.008
DyGKT	0.978	0.978	-0.000	0.934	0.934	0.000
DGEKT	0.981	0.981	-0.000	0.943	0.943	-0.000

Table 7: Train-only vs. full-log graph ablation: ASSISTments 2012.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.961	0.961	-0.000	0.907	0.907	-0.000
GIKT	0.967	0.968	0.000	0.915	0.915	0.000
SKT	0.953	0.953	0.000	0.898	0.898	0.000
DyGKT	0.952	0.952	-0.000	0.899	0.899	-0.000
DGEKT	0.960	0.960	0.000	0.907	0.907	0.000

5 DDR augmentation operator definitions

All operators act on the retained acyclic E_{pre} at perturbation strength $p \in \{0.05, 0.10, 0.20, 0.30\}$:

- **Attribute masking** masks a random subset of edge attributes without changing E_{pre} (DDR=0 by construction).
- **Edge dropping** removes each edge independently with probability p (expected DDR $\approx p$).

Table 8: Train-only vs. full-log graph ablation: XES3G5M.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.834	0.835	0.002	0.842	0.842	0.000
GIKT	0.878	0.878	0.000	0.860	0.860	0.000
SKT	0.625	0.624	-0.000	0.800	0.800	0.000
DyGKT	0.752	0.749	-0.003	0.813	0.812	-0.001
DGEKT	0.841	0.841	-0.000	0.844	0.844	-0.001

Table 9: Train-only vs. full-log graph ablation: Synthetic C2.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.873	0.873	-0.001	0.808	0.807	-0.001
GIKT	0.875	0.874	-0.000	0.809	0.809	0.000
SKT	0.708	0.708	0.000	0.722	0.723	0.001
DyGKT	0.710	0.710	0.000	0.726	0.726	-0.000
DGEKT	0.710	0.711	0.001	0.724	0.725	0.001

Table 10: Train-only vs. full-log graph ablation: Synthetic C5.

Model	AUC to	AUC fl	Δ AUC	ACC to	ACC fl	Δ ACC
GKT	0.810	0.812	0.002	0.738	0.738	0.000
GIKT	0.820	0.820	-0.000	0.749	0.747	-0.002
SKT	0.810	0.809	-0.001	0.740	0.739	-0.001
DyGKT	0.826	0.826	0.000	0.751	0.751	0.001
DGEKT	0.830	0.830	0.000	0.756	0.756	-0.001

- **Subgraph sampling** performs a random walk until $\lceil(1-p)|\mathcal{C}|\rceil$ nodes are visited and returns the induced subgraph (boundary edges drop; internal connectivity preserved).
- **Node dropping** removes $\lceil p|\mathcal{C}|\rceil$ nodes and all incident edges.
- **prereq_preserve** draws the same per-edge budget as edge dropping but removes only transitively redundant edges, protecting the transitive-reduction backbone.

6 Ground-truth disagreement examples (Junyi)

The cross-validation procedure exports qualitative disagreement cases to `results/tables/disagreement_examples.csv`. Inferred-only edges are predominantly bidirectional co-occurring arithmetic pairs (e.g. `number_line` $\leftrightarrow$ `addition_1`). Expert-only edges are systematic level-progression annotations (e.g. `equivalent_fractions_2` $\rightarrow$ `equivalent_fractions`). The `wrong_direction` category contains many reversed level-progression pairs such as `addition_1` $\rightarrow$ `addition_2`.

Table 11: Paired comparisons against *simpleKT* on public benchmarks (three-fold learner CV). Δ is mean AUC difference (model minus *simpleKT*); fold-level magnitudes in the main baseline-CV table are the primary evidence. p_t and p_W are reported for transparency only (underpowered at $n=3$ folds).

Dataset	Model	Δ AUC	p_t	p_W
ASSISTments 2012	GKT	-0.0069	<0.001	0.25
	GIKT	-0.0005	0.001	0.25
	SKT	-0.0154	<0.001	0.25
	DyGKT	-0.0160	<0.001	0.25
	DGEKT	-0.0079	<0.001	0.25
XES3G5M	GKT	-0.0411	<0.001	0.25
	GIKT	+0.0031	<0.001	0.25
	SKT	-0.2501	<0.001	0.25
	DyGKT	-0.1223	<0.001	0.25
	DGEKT	-0.0334	<0.001	0.25
Junyi Academy	GIKT	-0.0021	0.445	1.00
	SKT	-0.0055	0.031	0.25
	DyGKT	-0.0066	0.100	0.25
	DGEKT	-0.0034	0.267	0.25

Table 12: Sequence-autocorrelation diagnostics on the three public benchmarks. *KC rep.* and *Item rep.* are the fractions of consecutive within-learner interactions that repeat the previous KC or item; *Persist. acc.* is $P(y_t=y_{t-1})$; *Lag-1 ρ* is the lag-one label autocorrelation; *Prev-AUC* is the AUC of the trivial predictor $\hat{y}_t=y_{t-1}$.

Dataset	KC rep.	Item rep.	Persist. acc.	Lag-1 ρ	Prev-AUC
Junyi Academy	0.755	0.755	0.786	0.244	0.622
ASSISTments 2012	0.662	0.000	0.664	0.207	0.604
XES3G5M	0.211	0.145	0.786	0.342	0.671

7 Ground-truth metrics with node Jaccard

Table S14 (Table 13) extends the main manuscript ground-truth table with node Jaccard across the same top- K cutoffs.

8 Cold-start tables (full)

Table S12 (Table 14) reports fold 0 *simpleKT* metrics by KC-frequency stratum; Table S13 (Table 15) lists three-fold per-model AUC (mean $\pm$ std) and $\Delta_{\max}$ by stratum. The main manuscript summarises both in a single table.

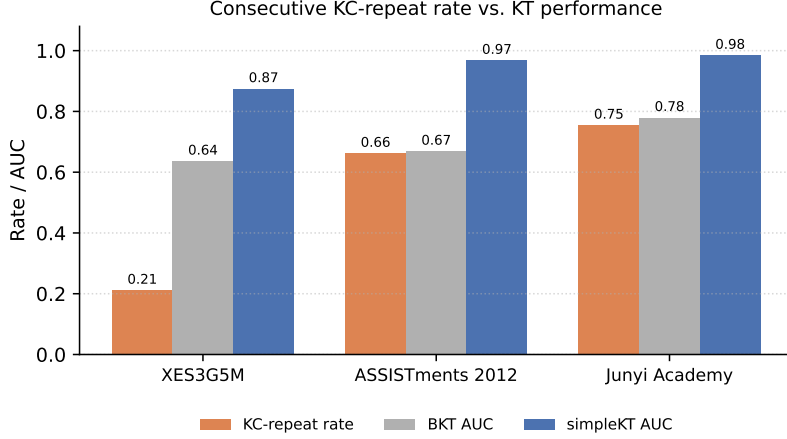


Fig. 1: Consecutive KC-repeat rate against *simpleKT* AUC across the three public benchmarks.

Table 13: Ground-truth cross-validation on Junyi: edge/direction/reachability metrics and node Jaccard across top- K cutoffs. The main manuscript table reports edge precision/recall, direction agreement, and reachability F1 only.

K	Edge F1	Dir. agr.	Rch. F1	Node Jaccard
200	0.071	0.346	0.008	0.290
500	0.109	0.225	0.021	0.453
800	0.150	0.280	0.062	0.572
$ \text{expert} =1131$	0.183	0.261	0.104	0.632
1200	0.185	0.262	0.126	0.641
1800	0.188	0.278	0.260	0.644
2500	0.187	0.285	0.339	0.610
5000 (cap)	0.187	0.285	0.339	0.610

9 DDR downstream scatter plot

10 Additional KT-graph figures

11 Training parity

Table S15 (Table 16) outlines hyperparameter parity and model configuration across *pyKT* and native baselines. Settings shared by all models are kept out of the table: learning rate 0.001 (Adam), zero hyperparameter-search budget, split seed 42, identical processed splits under both train-only and full-log graph constructions, and checkpoint selection by best validation AUC with early stopping (BKT is fit by L-BFGS on NLL without early stopping). Epoch caps and batch sizes differ across families (30 epochs for

Table 14: Per-stratum cold-start KC diagnostic with *simpleKT* on fold 0. The n column counts test interactions whose KC falls in the stratum train-frequency bin; *out of range* marks test KCs outside the four canonical bins. Stratum mass varies by dataset split; under default bins Synthetic C2/C5 route all test interactions into the hot stratum (> 500 train-fold hits per knowledge component).

Dataset	Stratum	n	AUC	ACC	NLL
Junyi Academy	very_cold	19	1.000	0.684	0.456
	cold	256	0.447	0.621	0.633
	warm	11,257	0.615	0.727	0.572
	hot	7,725,588	0.718	0.831	0.413
	out of range	10	0.500	0.900	0.511
ASSIST'12	very_cold	49	0.608	0.755	0.631
	cold	707	0.769	0.717	0.565
	warm	9,017	0.765	0.687	0.587
	hot	807,178	0.693	0.704	0.574
XES3G5M	very_cold	161	0.810	0.814	0.424
	cold	2,592	0.798	0.794	0.453
	warm	17,655	0.790	0.805	0.431
	hot	1,902,424	0.717	0.796	0.467
	out of range	6	0.625	0.667	0.638
Synthetic C2	hot	60,000	0.624	0.683	0.607
Synthetic C5	hot	60,000	0.630	0.611	0.651

sequence checkpoints vs. 10 for graph backbones in the primary release), reflecting per-epoch cost. Table S21 reports an epoch-matched GKT ablation (30 epochs, batch 64) on the aligned split seed 42. “Native” denotes our re-implementations in `src/models`; graph models consume the exported E_{pre} and, where available, E_{sim} edges.

12 ΔAUC intervals for primary pairs

Table S16 (Table 17) reports the mean AUC difference (ΔAUC) between the primary model pairs on XES3G5M under train-only graph construction. The released bundle reports paired- t 95% intervals over three learner-disjoint folds (default; matches fold-level ΔAUC). Optional row-level learner bootstrap: `scripts/bootstrap_auc_ci.py--learner-bootstrap`. The GKT deficit under the released 10-epoch budget (-0.041 ; CI $[-0.044, -0.038]$) is more than an order of magnitude larger than the interval half-width, so the model-specific conclusion does not hinge on fragile three-fold t -tests. Table S21 shows epoch matching recovers only $\approx +0.003$ mean AUC, leaving $\Delta \approx -0.038$ under matched caps. Reproducers may optionally run `python scripts/bootstrap_auc_ci.py --learner-bootstrap` on exported prediction parquets for row-level learner resampling (slower; fold-level paired- t is the default).

Table 15: Cold-start comparison of sequence-only (No-Graph) and graph-aware (Graph) KT models. Entries are three-fold mean $\pm$ std AUC by KC train-frequency stratum; $\Delta_{\max}$ is the best Graph AUC minus the best No-Graph AUC in the same dataset/stratum.

Dataset	Stratum	No-Graph		Graph			
		DKT	SimpleKT	GKT	GIKT	DGEKT	$\Delta_{\max}$
assist2012	very_cold	0.723 $\pm$ 0.085	0.724 $\pm$ 0.100	0.695 $\pm$ 0.100	0.726 $\pm$ 0.102	0.724 $\pm$ 0.100	+0.002 $\pm$ 0.143
	cold	0.769 $\pm$ 0.004	0.772 $\pm$ 0.003	0.750 $\pm$ 0.013	0.772 $\pm$ 0.003	0.772 $\pm$ 0.003	+0.000 $\pm$ 0.013
	warm	0.751 $\pm$ 0.016	0.756 $\pm$ 0.017	0.724 $\pm$ 0.015	0.757 $\pm$ 0.017	0.756 $\pm$ 0.017	+0.001 $\pm$ 0.024
	hot	0.687 $\pm$ 0.002	0.696 $\pm$ 0.002	0.601 $\pm$ 0.001	0.691 $\pm$ 0.002	0.687 $\pm$ 0.002	-0.005 $\pm$ 0.003
junyi	very_cold	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	0.813 $\pm$ 0.324	+0.000 $\pm$ 0.458
	cold	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	0.505 $\pm$ 0.239	+0.000 $\pm$ 0.338
	warm	0.631 $\pm$ 0.014	0.631 $\pm$ 0.014	0.626 $\pm$ 0.014	0.627 $\pm$ 0.014	0.626 $\pm$ 0.014	-0.004 $\pm$ 0.020
	hot	0.719 $\pm$ 0.000	0.719 $\pm$ 0.000	0.704 $\pm$ 0.001	0.713 $\pm$ 0.001	0.711 $\pm$ 0.001	-0.006 $\pm$ 0.001
synthetic_c2	hot	0.619 $\pm$ 0.002	0.626 $\pm$ 0.002	0.554 $\pm$ 0.000	0.634 $\pm$ 0.002	0.634 $\pm$ 0.002	+0.008 $\pm$ 0.003
synthetic_c5	hot	0.628 $\pm$ 0.003	0.630 $\pm$ 0.003	0.524 $\pm$ 0.003	0.633 $\pm$ 0.002	0.633 $\pm$ 0.002	+0.002 $\pm$ 0.005
xes3g5m	very_cold	0.820 $\pm$ 0.012	0.817 $\pm$ 0.013	0.826 $\pm$ 0.004	0.818 $\pm$ 0.013	0.818 $\pm$ 0.013	+0.005 $\pm$ 0.018
	cold	0.802 $\pm$ 0.006	0.804 $\pm$ 0.007	0.793 $\pm$ 0.007	0.804 $\pm$ 0.008	0.803 $\pm$ 0.008	-0.000 $\pm$ 0.010
	warm	0.783 $\pm$ 0.003	0.787 $\pm$ 0.003	0.745 $\pm$ 0.004	0.780 $\pm$ 0.003	0.776 $\pm$ 0.003	-0.007 $\pm$ 0.005
	hot	0.708 $\pm$ 0.000	0.717 $\pm$ 0.000	0.606 $\pm$ 0.000	0.709 $\pm$ 0.000	0.703 $\pm$ 0.000	-0.009 $\pm$ 0.000

13 Epoch-matched GKT ablation

Table S21 (Table 18) compares the released GKT configuration (10 epochs, batch 16) with an epoch-matched rerun (`configs/xes3g5m_gkt_epochs30.yaml`; 30 epochs, batch 64) on XES3G5M under train-only graphs and the aligned split seed 42. Fold-level Δ AUC intervals use paired- t over three learner-disjoint folds, matching Table S16. Multi-seed replication requires rebuilding fold-specific graph exports for each split base seed before retraining GKT.

14 Controlled Leak Injection

Table S17 (Table 19) probes the sensitivity of the audit metrics by deliberately injecting 5% and 20% of test-fold interactions into the train-only prerequisite builder on XES3G5M (fold 0). Edge counts $|E_{\text{pre}}|$ are *candidate* edges before DAG pruning (the main-text DAG audit reports 701 retained edges after pruning on the clean fold). Candidate-edge count and TBMR_f increase monotonically with the injection rate,

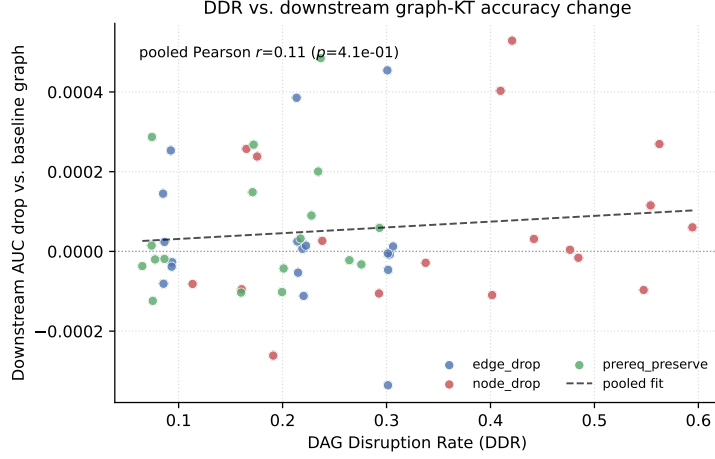


Fig. 2: Supplementary Figure S3. Downstream DGEKT sensitivity to prerequisite-graph disruption on ASSISTments 2012 and XES3G5M. Each point is one fold/operator/strength run; the vertical axis is AUC drop relative to the unperturbed train-only graph.

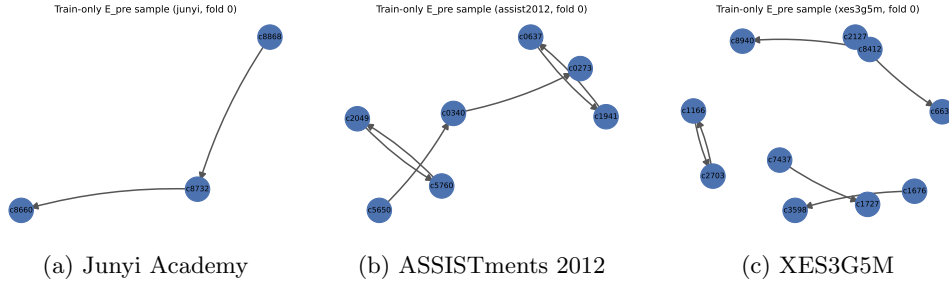


Fig. 3: Supplementary Figure S1. High-weight samples from train-only E_{pre} after DAG audit (fold 0 exports). The main manuscript shows the Junyi panel only.

while $\text{ECR}_f^{\text{flag}}$ remains 0 (learner-id sets stay disjoint) and $\text{ECR}_f^{\text{overlap}}$ is already saturated at 1.0 on this transition-dense corpus. Flag-level checks alone therefore miss this contamination channel, supporting the tiered audit design in the main manuscript.

Table S18 (Table 20) reports fold-0 test AUC after retraining *simpleKT*, GKT, and GIKT on the clean and injected graphs. Headline AUC does not track the structural audit monotonically: GKT and GIKT AUC *rise* at 20% injection on this dense log, whereas *simpleKT* remains near 0.850–0.858. We therefore treat S18 as evidence that metric-tier reporting must precede AUC-only leakage conclusions, not as proof that contamination always lowers accuracy.

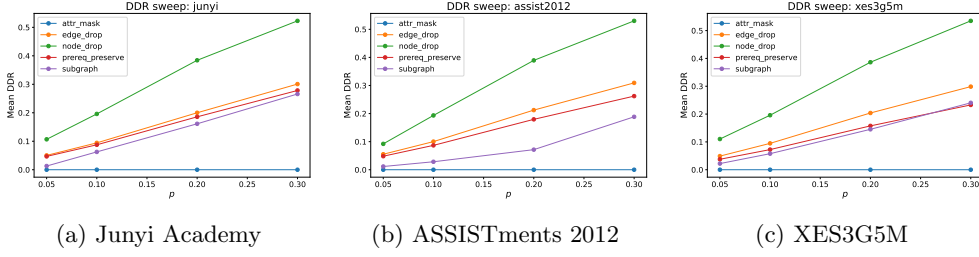


Fig. 4: Supplementary Figure S2. DDR versus perturbation strength p for five augmentation families across three benchmarks. The main manuscript shows the Junyi panel only.

Table 16: Training parity and hyperparameters (Table S15)

Model	Codebase	Graph input	Epochs	Batch	ES patience	Runtime	Hardware
BKT	scipy L-BFGS	—	30	64	N/A	<5 min	CPU
DKT	pyKT stock	—	30	64	5	30–180 min	A100/V100
<i>simpleKT</i>	pyKT stock	—	30	64	5	30–180 min	A100/V100
AKT	pyKT stock	—	30	64	5	30–180 min	A100/V100
GKT	pyKT stock	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
GIKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
SKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
DyGKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100
DGEKT	native	$E_{\text{pre}} + E_{\text{sim}}$	10	16	5	30–180 min	A100/V100

Epoch-matched GKT ablation: `configs/xes3g5m.gkt-epochs30.yaml` (30 epochs, batch 64).

Table 17: ΔAUC intervals for primary pairs, XES3G5M, train-only graphs (Table S16)

Model Pair	ΔAUC	95% CI (paired t , 3 folds)	n_{learners}	n_{rows}
GKT vs <i>simpleKT</i>	−0.041	[−0.044, −0.038]	—	—
GIKT vs <i>simpleKT</i>	+0.003	[+0.003, +0.003]	—	—

15 Exploratory ANOVA summaries

The main manuscript reports fold-level means $\pm$ std throughout Tables 7, 8, 5, 10, and 12, with Table S16 ΔAUC intervals (Table S16) as the primary inferential evidence for

Table 18: Epoch-matched GKT ablation on XES3G5M, split seed 42 (Table S21)

GKT config	Epochs	Batch	Mean AUC	Δ vs <i>simpleKT</i>	95% CI (3 folds)
Primary release	10	16	0.834	-0.041	$[-0.044, -0.038]$
Epoch-matched ablation	30	64	0.837	-0.038	$[-0.048, -0.027]$
<i>simpleKT</i> reference	30	64	0.875	—	—
Epoch gain (30ep - 10ep GKT)			+0.003		$[-0.004, +0.011]$

Table 19: Controlled Leak Injection, XES3G5M fold 0. $|E_{\text{pre}}|$ counts candidate edges before DAG pruning. (Table S17)

Injection rate	$ E_{\text{pre}} $	ECR ^{overlap}	ECR ^{flag}	TBMR
0%	1162	1.000	0.000	0.754
5%	1378	1.000	0.000	0.776
20%	1417	1.000	0.000	0.779

Table 20: Downstream test AUC under controlled leak injection, XES3G5M fold 0 (Table S18)

Model	Clean	5% leak	20% leak
<i>simpleKT</i>	0.850	0.852	0.858
GKT	0.810	0.825	0.860
GIKT	0.852	0.860	0.880

the XES3G5M ranking observation. For reviewer-requested completeness, Tables S19–S20 add exploratory ANOVA models on fold-level observations. Because each dataset–model cell contains only three folds, we interpret ANOVA p -values as descriptive rather than as proof of backbone dominance.

Table 21: Exploratory two-way ANOVA on fold-level AUC ($n=3$ folds per dataset-model cell; public benchmarks only, train-only graphs; Table S19). Partial $\eta^2 = \text{SS}_{\text{effect}}/(\text{SS}_{\text{effect}} + \text{SS}_{\text{error}})$. With only three folds per cell, p -values are underpowered; fold-level ΔAUC magnitudes and Table S16 intervals are the primary evidence.

Source	SS	df	MS	p	η_p^2
Dataset	0.3365	2	0.1682	<0.001	1.000
Backbone	0.0701	5	0.0140	<0.001	0.998
Dataset $\times$ Backbone	0.0968	10	0.0097	<0.001	0.999
Residual	0.0001	34	0.0000	—	—

Table 22: Exploratory ANOVA on DGEKT downstream AUC under graph perturbation (ASSISTments and XES3G5M; three folds per operator/strength; Table S20). Dependent variable: test AUC after retraining on the perturbed graph. Partial η^2 as above. This analysis supports the null predictive-sensitivity finding in Section 4.7; it does not establish backbone dominance.

Source	SS	df	MS	p	η_p^2
Dataset	0.1893	1	0.1893	<0.001	1.000
Operator	0.0000	2	0.0000	0.973	0.001
Operator $\times$ strength p	0.0000	6	0.0000	0.996	0.014
Dataset $\times$ Operator	0.0000	2	0.0000	0.890	0.006
Residual	0.0000	42	0.0000	—	—